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Microprocessor and Embedded Systems Laboratory

Final Project Report

Section: B2 Group: 01

DailyDesk Bangla- Clock Calendar with Notice features

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Academic Honesty Statement:

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"In signing this statement, We hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present), and cited all relevant sources while completing this project. We understand that if we fail to honor this agreement, We will each receive a score of ZERO for this project and be subject to failure of this course."

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2 Abstract

Our proposed project is a digital Bangla clock, time, and notice board that displays scrolling Bangla text, date, and time using an ESP32 module and an LED with a dot matrix display.
It connects to the internet to fetch real-time data (time and date) using an RTC (Real-Time Clock) module and stores it to operate offline. The display is divided into three LED screens, all showing synchronized information. The notices displayed here are dynamically updated via a website. The most challenging part of the entire project was to implement an algorithm for Bangla date, month, and year conversion, mapping the Bangla words in a dot matrix, and creating an IoT interface for a Bangla notice board. The system designed here is to be visually improved over older designs and is especially suitable for educational and agricultural institutions or organizations.

3 Introduction

In the socio-cultural and agricultural context of Bangladesh, daily activities, seasonal farming cycles, and traditional festivals are primarily structured around the Bangla calendar rather than the Gregorian (English) calendar. For instance, Notable occasions such as Asharh's monsoon planting season or the Nabanna harvest festival are determined based on specific dates in the Bangla calendar, which vary significantly from their Gregorian counterparts. The absence of native calendar integration in most electronic scheduling systems and displays poses a barrier to accessibility, especially for individuals in rural or semi-urban communities who may have limited familiarity with English.

The deployment of English-language date, time, and notice board systems using LED dot matrix displays is widely prevalent. However, similar implementations in Bangla script are comparatively rare. This project introduces a Bangla-language IoT-enabled digital notice board integrated with a calendar and clock system, utilizing LED dot matrix displays to render information in Bangla. The system is architected around the ESP32 microcontroller, which is responsible for simultaneously controlling three LED dot matrix modules.

Utilizing the ESP32's built-in Wi-Fi capabilities, the device establishes a connection to the internet, enabling it to function as a fully operational IoT system. It retrieves real-time date and time data through an internal RTC (Real-Time Clock) synchronization mechanism, ensuring accurate and periodic updates without manual intervention.The designed system is compact, easily mountable, and well-suited for indoor environments such as schools, offices, or public buildings.

To complete the design, a custom enclosure was developed to securely house all critical components, including the ESP32 module, three LED displays, and the regulated power supply, providing both functionality and mechanical safety. This enhances usability for non-English-speaking populations and aligns with local traditions and agrarian lifestyles, making the technology both inclusive and context-aware.

4 Design

4.1 Problem Formulation (PO(b))

4.1.1 Identification of Scope

Hardware Scope

- ESP32 microcontroller (WiFi)
- External RTC module: DS3231 RTC for accuracy
- MAX7219 Dot Matrix LED Display (32x8 for 3 type display)
- Push button for manual control: (optional)
- Power from 220V(ac) to operate ESP32

Software Scope

- NTP-based RTC syncing (with fallback to DS3231 if no WiFi)
- Conversion from English to Bangla numerals and dates
- Custom firmware for:
    - Time fetching and updating
    - Scrolling text display
    - Flash memory-based storage for notices

User Interface Scope

- Design a user-friendly interface that allows easy navigation and interaction with the notice board, clock, and calendar functionalities.
- Consider usability principles to ensure accessibility for users with varying levels of technical proficiency.

4.1.2 Literature Review

While most commercial devices use the Gregorian calendar, research has shown the importance of localizing interfaces to include native languages and cultural calendars, such as the Bengali calendar (Bangla Barshabda). Few open-source works attempt to convert Gregorian to Bangla dates algorithmically, referencing systems used by Prothom Alo and Bangla calendar apps. These rely on approximate seasonal mapping and leap year adjustments. Integration of Bangla numerals and text rendering into displays, especially limited ones like dot matrix displays, remains a technical challenge, with custom bitmap fonts often required.
We generate initial ideas from *S. A. Khan, S. T. Azad, T. Mondal, A. J. Bin Iqbal and S. M. Choudhury, "Development of an Internet of Things based Bangla Calendar Clock," 2023 26th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 2023,* and further tried to develop and add new features on the previous design with minimized budget.

4.1.3 Formulation of Problem

In today's digital age, while numerous clock and calendar solutions exist, localized, culturally-relevant embedded systems are largely missing from personal or institutional desktop utilities. Most off-the-shelf digital clocks display only the Gregorian calendar and time in English, offering no support for the Bangla language, Bangla numerals, or the Bangla calendar system. Furthermore, traditional wall clocks or calendars do not support dynamic updates or digital notices, which are essential for daily planning and real-time communication in study desks, dormitories, or small office spaces.

4.1.4 Analysis

Metric	Target
Time Accuracy	±5 seconds/month (with DS3231), updated in 2 hours
Power Consumption	<100 mA in normal operation 220V(ac)>9V(dc)>Buck converter
Display Readability	Clear Bangla numerals using dot matrix in big-endian column-major (32*8 display)
Storage Capacity	ESP32 is significantly more powerful, enabling smoother scrolling, multitasking, and storage for custom Bangla fonts compared to Arduino.
Bangla Date Accuracy	Matches with the official Bangla calendar (Bangladesh)

4.2 Design Method (PO(a))

We used an ESP32 board and connected it to the LED dot matrix and programmed the board to use it as a Bangla notice board and also show the date and time in Bangla. We configured the internal RTC module to update the date and time. The algorithm deployed in this project-
=>Connecting to WiFi via WiFi Manager.h library interface
=>Fetching date and updating time on RTC
=> Bitmapping Bangla characters using the dot matrix tool
=> Conversion of characters to Bangla using the MD_MAX72xx library
=> Interfacing a website to show desired notices in Bangla and Printing required data on screen using MD Parola library
Bitmapping:



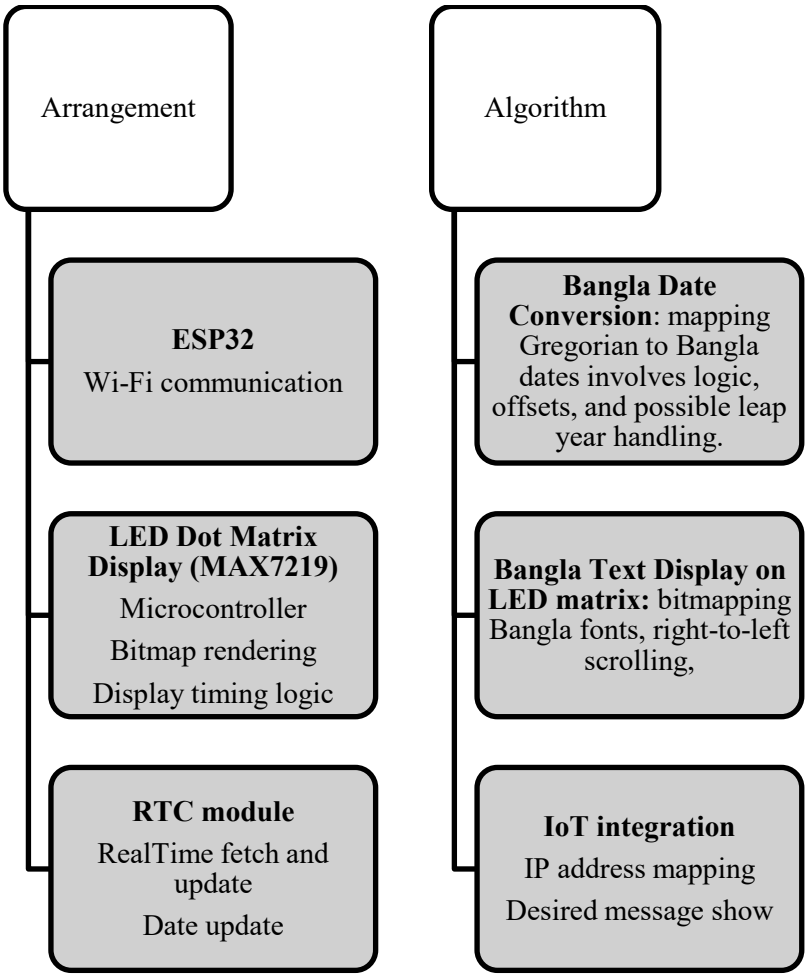
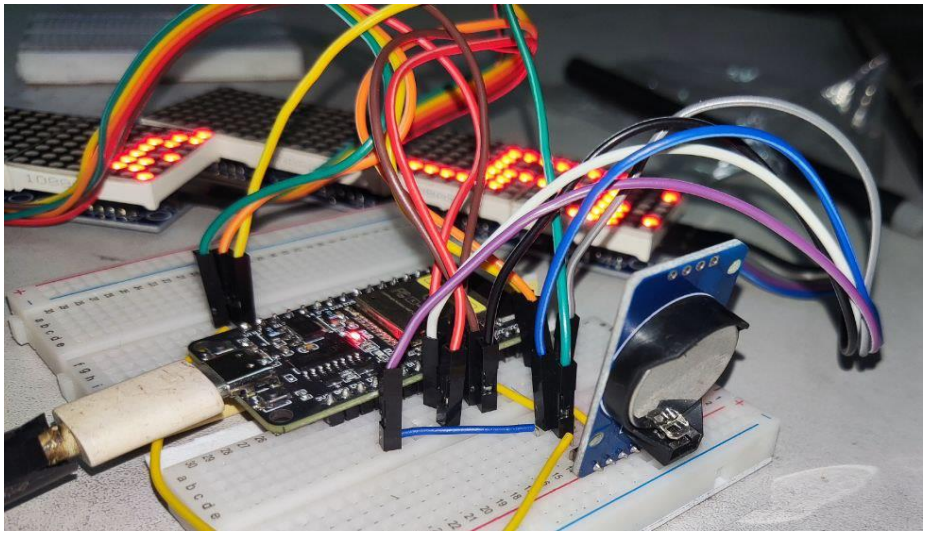
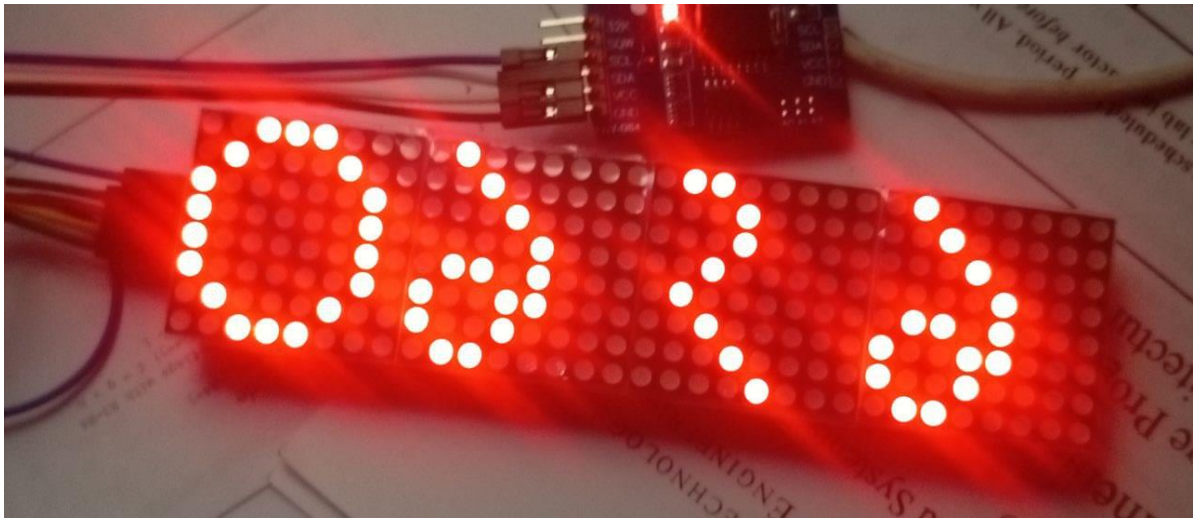
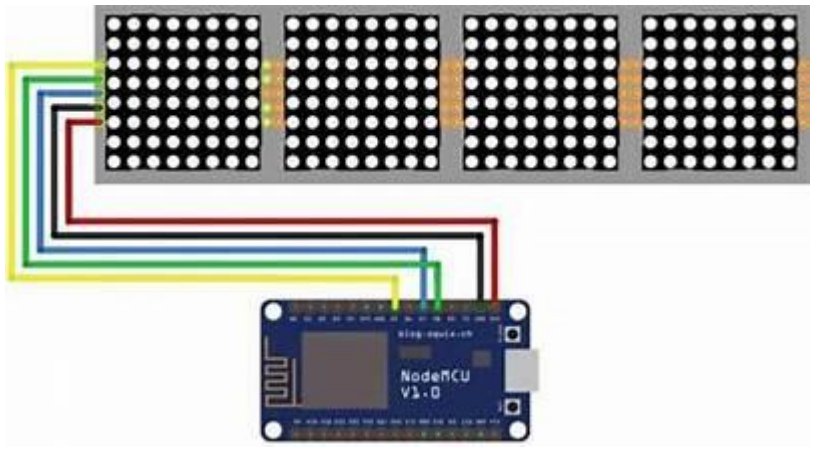


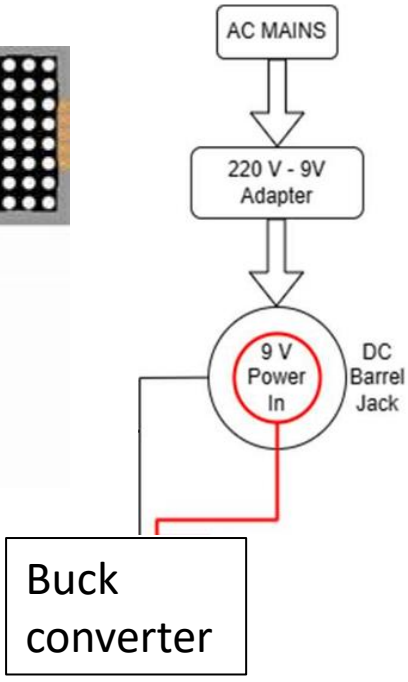
Fig: Design setup

4.3Circuit Diagram





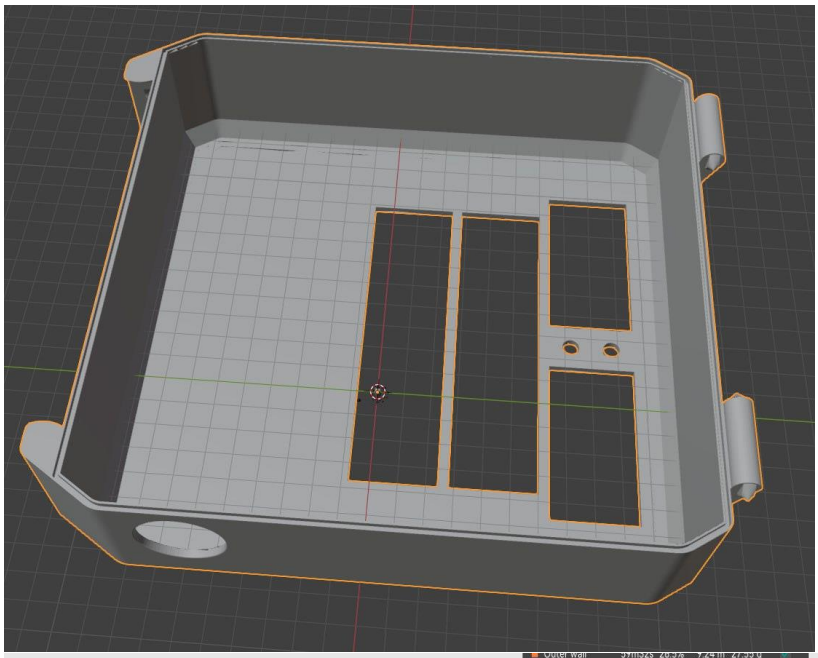


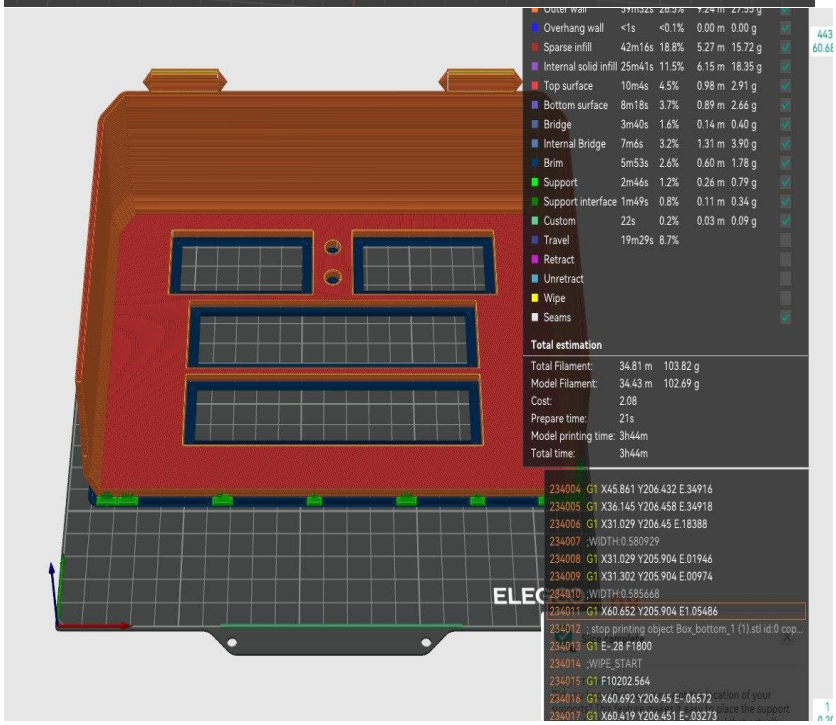


Buck converter

```
#define MAX_DEVICES     4
#define DIN_PIN         23
#define CLK_PIN          18
#define CS1_PIN           5  // Time
#define CS2_PIN          16  // Date
#define CS3_PIN          17  // Message
```

4.4 3D Rendering





4.5Full Source Code of Firmware

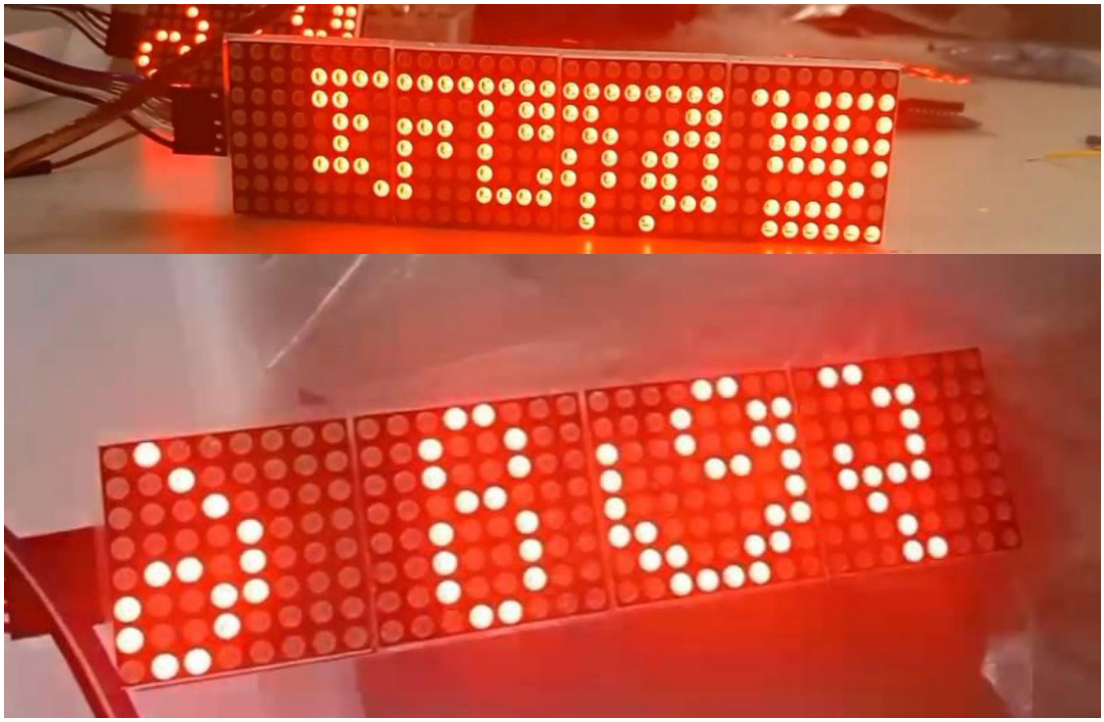
Added as an appendix

Table: Source Code for the main program

5 Implementation

5.1 Description





The practical considerations of the design are- Suitable for placement on both tables and walls, Affordable, could have been more cost-effective if a custom-made multi-segment display had been used. The designed website can be conveniently accessed using a smartphone

6 Design Analysis and Evaluation

6.1 Novelty

Most digital desk clocks display only the Gregorian calendar. This project incorporates the Bangla calendar system (month, date, year), which is highly significant in Bangladeshi culture, yet underrepresented in digital devices. Custom Bangla digits and letters are rendered on low-resolution matrix displays, which usually do not support complex scripts like Bangla. This requires bitmap-level font creation, making the display linguistically inclusive and culturally rich. Users can input custom notices that scroll across the display — a feature often found in large signage boards but rarely in compact, personal desk setups. The system smartly handles time fetching from both an RTC module (DS3231) and the internet (NTP over WiFi), allowing it to function offline and online, increasing reliability. Besides, previous works focused on Arduino or ESP8266 modules, which are slow, have less memory and control compared to our design with esp32.

6.2 Design Considerations (PO(c))

6.2.1 Considerations to public health and safety

The project uses low-voltage components (e.g., ESP32, DS3231 RTC, dot matrix LED), which are safe for public use, especially in indoor environments like school desks, office tables, or dormitories.The system avoids harmful radiation or noise emissions, ensuring no adverse effects on human health.The power supply is regulated and isolated, reducing the risk of electric shock or fire hazards.The display brightness is moderate and adjustable, which helps reduce eye strain during continuous viewing.

6.2.2 Considerations to environment

The system operates on low power (typically 3.3V–5V) and consumes very little energy, aligning with energy-efficient design principles. Most components used (e.g., ESP32, RTC, LED modules) are RoHS compliant, ensuring minimal environmental pollution from lead or hazardous materials.The device is reusable and modular. If any part fails (like the LED matrix or RTC), it can be replaced without discarding the entire unit, supporting e-waste reduction. The use of recyclable components and minimal plastic casing reduces its environmental footprint.

6.2.3 Considerations to cultural and societal needs

The inclusion of the Bangla calendar and Bangla font directly caters to local cultural needs, enhancing user engagement and making it meaningful to the Bangladeshi audience. In many rural or traditional settings, people are more familiar with the Bengali date system. This project bridges the gap between digital technology and local tradition. The notice board feature can be used in schools, colleges, and workplaces to display timely information in Bangla, improving communication and awareness in native language. It promotes digital literacy in local languages, encouraging more people, especially the elderly or less English-proficient individuals, to interact with digital displays.

6.3 Investigations (PO(d))

6.3.1 Data Collection

মাসের নাম	দিন (বাংলাদেশ, ২০১৯–বর্তমান)	ঐতিহ্যগত ঋতু বঙ্গে	মাসের নাম (গ্রেগরীয় বর্ষপঞ্জি)
বৈশাখ	৩১	গ্রীষ্ম	এপ্রিল–মে
জ্যৈষ্ঠ	৩১		মে, জুন
আষাঢ়	৩১	বর্ষা	জুন–জুলাই
শ্রাবণ	৩১		জুলাই–আগস্ট
ভাদ্র	৩১	শরৎ	আগস্ট–সেপ্টেম্বর
আশ্বিন	**৩১**		সেপ্টেম্বর–অক্টোবর
কার্তিক	৩০	হেমন্ত	অক্টোবর–নভেম্বর
অগ্রহায়ণ	৩০		নভেম্বর–ডিসেম্বর
পৌষ	৩০	শীত	ডিসেম্বর–জানুয়ারি
মাঘ	৩০		জানুয়ারি–ফেব্রুয়ারি
ফাল্গুন	**২৯/৩০ (অধিবর্ষ)**	বসন্ত	ফেব্রুয়ারী–মার্চ
চৈত্র	৩০		মার্চ–এপ্রিল

English Date/Month(2025-2026)	বাংলা মাস (১ তারিখ,1432)
14 April 2025	বৈশাখ
15 May 2025	জ্যৈষ্ঠ
15 June 2025	আষাঢ়
16 July 2025	শ্রাবণ
17 August 2025	ভাদ্র
17 September 2025	আশ্বিন
18 October 2025	কার্তিক
17 November 2025	অগ্রহায়ণ
17 December 2025	পৌষ
16 January 2026	মাঘ
15 February 2026	ফাল্গুন
15 March 2026	চৈত্র

The data collected falls under two main categories: temporal data (Bangla time and date) and user-generated content (Bangla notices/messages). The challenge was to convert the English date and time into Bangla date and time. The system is designed to operate as an IoT-enabled embedded device, primarily powered by the ESP32 microcontroller with Wi-Fi capabilities.

6.3.2 Results and Analysis

6.4Limitations of Tools (PO(e))

1. ESP32 Microcontroller: Voltage Sensitivity
2. DS3231 RTC Module: Lack of DST/Timezone Awareness: DS3231 provides raw timekeeping without automatic adjustment. Besides, it has a limited Interface: It supports only I2C communication, which might get congested if many I2C devices are used together.
3. Dot Matrix LED Display (e.g., MAX7219 based)
    - Limited Resolution: Low pixel density makes it hard to display detailed fonts or complex characters, especially in Bangla script.
    - Character Flicker: Poor handling of refresh rates can cause visible flickering if the display is not properly multiplexed.

6.5Impact Assessment (PO(f))

6.5.1 Assessment of Societal and Cultural Issues

The project contributes positively to society by offering a low-cost, localized digital notice and time display system. By incorporating Bangla language support, it directly addresses the cultural relevance and accessibility for Bangladeshi users, especially in educational institutions, offices, and public spaces. It promotes digital inclusion by making technological interfaces more intuitive for native speakers.

6.5.2 Assessment of Health and Safety Issues

The project primarily operates on low-voltage electronics, posing minimal electrical hazards if enclosed properly. Sharp-edged components and exposed wires must be insulated and safely mounted. Brightness levels of the LED matrix should be calibrated to avoid visual discomfort or eye strain. Overall, with proper casing and adherence to basic electrical safety norms, the system is safe for both developers and users

6.5.3 Assessment of Legal Issues

Since the system does not store or transmit personal or sensitive data, data protection and privacy concerns are minimal.

6.6 Sustainability Evaluation (PO(g))

The project ensures sustainability by using low-power components like the ESP32 microcontroller and LED dot matrix displays. It replaces traditional paper calendars and notice boards, reducing paper waste and promoting digital alternatives. The compact design minimizes material use and e-waste, while modular components allow easy maintenance and long-term use. Additionally, by supporting the Bangla calendar and script, the system fosters cultural sustainability and inclusivity, especially in rural areas.

6.7Ethical Issues (PO(h))

As the displayed content is general in nature—such as public notices, time, and calendar information—there is minimal risk of data misuse or privacy breaches. This enhances the system's reliability and makes it suitable for community-wide deployments without requiring complex security protocols.

7 Reflection on Individual and Team work (PO(i))

7.1Individual Contribution of Each Member

Team Member	Contribution
2006099	Literature Review, Hardware assembly, RTC module, Bangla Year conversion logic, IoT interface design
2006100	Overview, Bangla month, day bitmapping, Conversion algorithm, IoT interface design
2006101	Overview, Power supply, IoT interface design, Bangla number bitmapping
2006102	3D rendering, Hardware assembly, IoT interface design, Code debugging

7.2 Mode of TeamWork

- Our project embraced collaborative teamwork through regular brainstorming sessions, task sharing, and iterative feedback
- Team diversity in terms of technical expertise and perspectives ensured creative solutions, enhanced problem-solving, and an inclusive approach to design challenges.
- Clear communication, well-defined roles, accountability, and trust are the key drivers of our successful teamwork.

7.3Log Book of Project Implementation

Date	Task
Project finalization	14/05/2025
Project design	20/05/2025
Hardware planning	30/05/2025
Bitmapping	10/06/2025
Clock interface	25/06/2025
Hardware assemble	05/07/2025
Code tuning and bitmapping	10/07/2025
IoT interface and finalization	28/07/2025

8 Communication to External Stakeholders (PO(j))

8.1 Executive Summary

The DailyDesk Bangla – Clock Calendar with Notice Features is an IoT-based smart information display system designed to promote digital accessibility in native Bangla script. Built using the ESP32 microcontroller integrated with an external RTC module, the system provides real-time date, time, and notice updates in Bangla, making it highly relevant for local contexts such as agriculture, education, and rural communities. Unlike conventional English-only digital clocks, this system aligns with the Bangla calendar, which is widely followed in farming schedules, seasonal festivals, and academic events. The device connects to the internet via Wi-Fi to fetch updated date and time data from servers and displays them on LED dot matrix displays, along with custom notices.The system is energy-efficient, cost-effective, and culturally inclusive, aiming to bridge the digital divide by serving users unfamiliar with English. Its robust yet simple design allows easy installation in schools, local offices, agricultural centers, and public places, contributing to sustainable digital transformation in rural Bangladesh.

9 Project Management and Cost Analysis (PO(k))

Product	Quantity	Per unit Cost	Total Cost
ESP32	1	450	450
32*8 dot matrix display	3	350	1050
LM 2596	1	1	60
Jumper	---	---	---
Power adapter	1	140	140
RTC module	1	220	220
Socket	1	10	10
3D rendering	1	2000	2000
Total cost			3930/-

10 Future Work (PO(l))

By utilizing the software template, a custom clock can be designed in any native script worldwide. While the current implementation uses an ESP32, the system's size and efficiency can be further optimized for industrial applications by programming the ATMEGA328P microcontroller directly. Given its cultural relevance in Bangladesh, integrating the Arabic calendar alongside Bangla and English is highly feasible. Integration of sensors (e.g., temperature, humidity, CO_2) to display live environmental data along with time/calendar, making it useful for smart classrooms or offices and connecting with a bell or buzzer to automate class change notifications(educational institutions) or raining alert(agricultural institutions) along with the time display can uplift our design.

11 References

1. S. A. Khan, S. T. Azad, T. Mondal, A. J. Bin Iqbal and S. M. Choudhury, "Development of an Internet of Things based Bangla Calendar Clock," 2023 26th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh, 2023, pp. 1-6, doi: 10.1109/ICCIT60459.2023.10441436.
2. GitHub - sabbirhossainujjal/Bangla-Calendar-Clock
3. *https://dotmatrixtool.com/*
4. *https://bn.wikipedia.org/wiki/%E0%A6%AC%E0%A6%99%E0%A7%8D%E0%A6%97%E0%A6%BE%E0%A6%AC%E0%A7%8D%E0%A6%A6*

```
#include <Wire.h>
#include <WiFi.h>
#include <RTClib.h>
#include <SPI.h>
#include <MD_Parola.h>
#include <MD_MAX72xx.h>
#include <WiFiUdp.h>
#include <NTPClient.h>

// -- Wi-Fi credentials (fill in your SSID/password) --
const char*  WIFI_SSID     = "Microprocessor Lab";
const char*  WIFI_PASSWORD = "bueteee437";

// Web-server on port 80
WiFiServer server(80);

// The HTML page served to clients
const char WebPage[] PROGMEM = R"rawliteral(
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>Smart Notice Board</title>
  <style>
    body { font-family: Arial; padding: 20px; }
    input[type="text"], .preset-buttons button { width: 300px; font-size: 16px; }
    .radio-group, button, input[type="range"] { margin: 10px 0; }
    .preset-buttons button { display: block; margin: 5px 0; }
  </style>
  <script>
    function sendIndex(idx) {
      fetch('/setIndex?idx=' + idx);
    }
    function sendSpeed(v) {
      fetch('/setSpeed?speed=' + v).catch(console.error);
    }
    function sendData() {
      let msg = encodeURIComponent(document.getElementById('message').value);
      fetch('/setMessage?msg=' + msg).catch(console.error);
    }
    window.addEventListener('load', () => {
      document.getElementById('speed')
        .addEventListener('input', e => sendSpeed(e.target.value));
      document.getElementById('sendBtn')
        .addEventListener('click', sendData);
    });
  </script>
</head>
<body>
  <h2>Smart Notice Board</h2>
  <label><b>Message:</b></label><br>
  <input type="text" id="message" maxlength="255"><br>
  <button id="sendBtn">Send Data</button>

  <div class="radio-group">
    <input type="radio" name="Invert" value="0" checked> Normal
    <input type="radio" name="Invert" value="1"> Inverse
  </div>
  <div class="radio-group">
    <input type="radio" name="ScrollType" value="L" checked> Left Scroll
    <input type="radio" name="ScrollType" value="R"> Right Scroll
  </div>

  <label><b>Speed:</b></label><br>
  <input type="range" id="speed" min="10" max="200" value="150"><br>

<small>Fast                 
;                  &nb
sp;Slow</small><br><br>

  <div class="preset-buttons">
    <button onclick="sendIndex(0)">সিটি নেই</button>
    <button onclick="sendIndex(1)">ক্লাস শেষ</button>
    <button onclick="sendIndex(2)">ফজর শুরু</button>
    <button onclick="sendIndex(3)">সিটি হবে</button>
    <button onclick="sendIndex(4)">কাল বন্ধ</button>
    <button onclick="sendIndex(5)">ধন্যবাদ</button>
  </div>
</body>
</html>
)rawliteral";

// NTP (UTC) + offset for UTC+6 (in seconds)
```

```
WiFiUDP ntpUDP;
NTPClient timeClient(ntpUDP, "pool.ntp.org", 6 * 3600);

// how often to re-sync RTC (ms)
const unsigned long NTP_RESYNC_INTERVAL = 2UL * 60UL * 60UL * 1000UL;
unsigned long lastNtpSync = 0;

// Hardware configuration
#define HARDWARE_TYPE   MD_MAX72XX::FC16_HW
#define MAX_DEVICES     4
#define DIN_PIN         23
#define CLK_PIN         18
#define CS1_PIN          5  // Time
#define CS2_PIN         16  // Date
#define CS3_PIN         17  // Message

static const uint8_t MODULES       = MAX_DEVICES;
static const uint8_t DISPLAY_WIDTH = MODULES * 8;

// Create matrix instances
MD_Parola timeDisplay = MD_Parola(HARDWARE_TYPE, DIN_PIN, CLK_PIN, CS1_PIN, MAX_DEVICES);
MD_Parola dateDisplay = MD_Parola(HARDWARE_TYPE, DIN_PIN, CLK_PIN, CS2_PIN, MAX_DEVICES);
MD_Parola msgDisplay  = MD_Parola(HARDWARE_TYPE, DIN_PIN, CLK_PIN, CS3_PIN, MAX_DEVICES);

RTC_DS3231 rtc;

// ──────────────────────────────────────────────
// Application state
bool    useCustomMessage = false;
String  customMessage;
int     selectedMessage = 0;

// Message scrolling state
int           msgScrollPos      = 0;
unsigned long msgLastScroll     = 0;
unsigned long msgScrollInterval = 150;  // adjustable via slider

// ──────────────────────────────────────────────
// Bangla date representation
struct BanglaDate {
  int day, month, year;
};

// Bangla digit bitmaps
const uint8_t banglaDigits[10][8] = {
  {0x00,0x1c,0x22,0x41,0x41,0x41,0x22,0x1c}, // ০
  {0x06,0x89,0x49,0x26,0x1c,0x00,0x00,0x00}, // ১
  {0x00,0x98,0x8c,0x4b,0x79,0x00,0x00,0x00}, // ২
  {0x3c,0x06,0x03,0x61,0x91,0x93,0x66,0x7c}, // ৩
  {0x00,0x00,0x6e,0x91,0x91,0x6e,0x00,0x00}, // ৪
  {0x3c,0x42,0x81,0x41,0x39,0x25,0x02,0x00}, // ৫
  {0x1c,0x02,0x01,0xf9,0x11,0x22,0x1c,0x00}, // ৬
  {0x00,0x00,0x60,0x90,0x90,0x7f,0x00,0x00}, // ৭
  {0x00,0x7e,0x09,0x19,0x2a,0x44,0x00,0x00}, // ৮
  {0x00,0x87,0x8b,0x88,0x88,0x47,0x22,0x1e}  // ৯
};

// Bangla month names (32-col each)
const uint8_t banglaMonthsBitmaps[12][64] = {

{0x20,0xac,0xfe,0x73,0x21,0x2c,0x2e,0x32,0x33,0x3f,0x20,0x24,0x24,0x3c,0x2c,0x24,0x20,0x3f,0x20,0x3f,0x20,0x1c,0
x24,0x04,0x3c,0x20,0x26,0x3e,0x02,0x02,0x3f,0x20}, // বৈশাখ

{0x20,0xae,0x7e,0x73,0x21,0x2c,0x22,0x23,0x39,0x3d,0x35,0x36,0x38,0x2b,0x2e,0x24,0x20,0x36,0x3b,0x20,0x20,0x20,0
x3e,0x2a,0x2b,0xbf,0x61,0x61,0x31,0x3f,0x2e,0x20}, // জ্যৈষ্ঠ

{0x7c,0x46,0x43,0x71,0x71,0x61,0x63,0x7e,0x42,0x42,0x7f,0x40,0x7f,0x40,0x63,0x55,0x5d,0x49,0x49,0x49,0x7f,0x40,0
x7f,0x40,0x7c,0x44,0x44,0x45,0x55,0x54,0x5c,0x40}, // আষাঢ়

{0x00,0x40,0x46,0x41,0x5d,0x3d,0x1d,0x7d,0x43,0x43,0x7f,0x40,0x40,0x40,0x40,0x7f,0x40,0x44,0x4c,0x54,0x66,0x7f,0
x00,0x18,0x30,0x20,0x30,0x7f,0x40,0x40,0x00,0x00}, // শ্রাবণ

{0x00,0x00,0x80,0xf0,0x9c,0x86,0x82,0xf2,0xf2,0xd6,0x9c,0x80,0x80,0xfe,0x80,0x86,0x83,0x81,0xf9,0x9a,0x92,0x92,0
x9f,0x80,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}, // ভাদ্র

{0x00,0x40,0x5c,0x44,0x44,0x6c,0x78,0x48,0x4c,0x7e,0x40,0x7e,0x60,0x3e,0x20,0xfe,0xc0,0xa0,0xb8,0x9a,0xbd,0xa5,0
xff,0xc0,0x80,0xb8,0xa8,0x20,0x7e,0x40,0x40,0x00}, // আশ্বিন

{0x20,0x26,0x2e,0x3a,0x33,0x3f,0x30,0x3a,0x2e,0x20,0x3f,0x20,0x3f,0x20,0xff,0xa1,0xa1,0xb9,0xb9,0x3f,0x60,0xa6,0
x2e,0x3a,0x33,0x3f,0x30,0x3a,0x2e,0x20,0x20,0x20}, // কার্তিক
```

```
{0xff,0x81,0x81,0xe1,0xe1,0xff,0x82,0xff,0x80,0x01,0x6b,0x9a,0x42,0xff,0x80,0xec,0xce,0xea,0x93,0x81,0x80,0xff,0
x80,0xc5,0xec,0x9e,0x83,0xff,0x80,0x8c,0x88,0xff}, // অগ্রহায়ণ

{0x16,0x1f,0x19,0x1b,0x13,0x00,0x12,0xde,0x92,0x96,0x8c,0x90,0x9f,0xd0,0x7f,0x10,0x1b,0x1b,0x1f,0x15,0x15,0x1f,0
x10,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00,0x00}, // পৌষ

{0x00,0x00,0x00,0x80,0x80,0x90,0xb7,0xfd,0xc9,0x8f,0x88,0x88,0x8c,0xff,0x80,0xff,0x80,0x80,0xf2,0xd6,0xde,0x9b,0
x81,0x81,0xff,0x80,0x80,0x80,0x80,0x80,0x00,0x00}, // মাঘ

{0x80,0xc0,0xc6,0xae,0x9a,0x81,0x81,0xbf,0xa0,0xac,0xbc,0x80,0xff,0x80,0x00,0xd3,0xd1,0xf1,0x85,0x8d,0xfb,0x80,0
x80,0x8c,0x9c,0x90,0x90,0x88,0xff,0x80,0x00,0x00}, // ফাল্গুন

{0x80,0x80,0x80,0xcc,0xd2,0xe1,0x41,0x40,0x7e,0x49,0x49,0x4a,0x44,0x40,0x40,0x46,0x41,0x41,0x42,0x44,0x74,0x52,0
x7f,0x40,0x40,0x40,0x00,0x00,0x00,0x00,0x00,0x00}  // চৈত্র
};

// Bangla weekdays (32-col each)
const uint8_t banglaWeekdaysBitmaps[7][64] = {

{0x40,0x40,0x5c,0x7e,0x62,0x43,0x61,0x60,0x72,0x52,0x56,0x5c,0x44,0x44,0x46,0x7f,0x40,0x40,0x7f,0x40,0x60,0x6e,0
x7f,0x59,0x4f,0x44,0x44,0x44,0x7f,0x40,0x40,0x40}, // রবিবার

{0x40,0x40,0x5c,0x7e,0x62,0x43,0x61,0x60,0x72,0x52,0x56,0x5c,0x44,0x44,0x46,0x7f,0x40,0x40,0x7f,0x40,0x60,0x6e,0
x7f,0x59,0x4f,0x44,0x44,0x44,0x7f,0x40,0x40,0x40}, // সোমবার

{0x60,0x66,0x7f,0x79,0x49,0x4f,0x44,0x44,0x7f,0x40,0x5c,0x44,0x74,0x7c,0x50,0x52,0x7a,0x6e,0x60,0x60,0x7f,0x40,0
x40,0x4f,0x49,0x49,0x4c,0x48,0x48,0x4f,0x7f,0x40}, // মঙ্গলবার

{0x40,0x40,0x40,0x40,0x40,0x58,0x58,0x5f,0x65,0x65,0x65,0x65,0x67,0x67,0x7f,0x44,0x44,0x44,0x00,0xe0,0xfe,0x9e,0
x12,0x12,0x12,0x12,0x72,0x7e,0x40,0x40,0x40,0x40}, // বুধবার

{0x40,0x40,0x40,0x40,0x40,0x58,0x58,0x5f,0x65,0x65,0x65,0x65,0x67,0x67,0x7f,0x44,0x44,0x44,0x00,0xe0,0xfe,0x9e,0
x12,0x12,0x12,0x12,0x72,0x7e,0x40,0x40,0x40,0x40}, // বৃহস্পতিবার

{0x40,0x40,0x78,0x68,0x78,0x30,0x78,0x6b,0x7b,0x43,0x43,0x7e,0x42,0x42,0x42,0x40,0x4c,0x46,0x52,0x5b,0x69,0x65,0
x7f,0x40,0x58,0x78,0x40,0x40,0x40,0x40,0x00,0x00}, // শুক্রবার

{0x00,0x20,0x20,0x20,0x2d,0x2d,0x2c,0x32,0x32,0x32,0x32,0x33,0x3f,0x20,0x20,0x20,0xff,0xa0,0xa0,0xac,0xbe,0xb2,0
xb2,0xb2,0xb3,0xbf,0xa0,0xa0,0x20,0x00,0x00,0x00}  // শনিবার
};

// Bangla messages
const uint8_t banglaMessagesBitmaps[][64] = {
  {0x20, 0xff, 0xa0, 0xb1, 0xb3, 0xae, 0xa4, 0x26, 0xbf, 0xa0, 0xff, 0xa0, 0xa0, 0xff, 0xa1, 0xa5, 0x07, 0x00,
0x2f, 0x37, 0xa5, 0xa7, 0xa0, 0xae, 0xac, 0xbf, 0xa0, 0xb2, 0xf6, 0x36, 0x3b, 0x01}, // ct nei
  {0x41, 0x4e, 0x5d, 0x76, 0x66, 0x7f, 0x60, 0x70, 0x40, 0x7e, 0x40, 0x66, 0x74, 0x5c, 0x44, 0x7e, 0x00, 0x1c,
0x76, 0x4a, 0x4e, 0x70, 0x70, 0x30, 0x7e, 0x40, 0x64, 0x7e, 0x5a, 0x4a, 0x7e, 0x00}, // class shesh
  {0x00, 0x64, 0x7c, 0x54, 0x42, 0x5e, 0x50, 0x58, 0x40, 0x7e, 0x42, 0x72, 0x76, 0x6c, 0x73, 0x5c, 0x40, 0x4d,
0x5e, 0x72, 0x7e, 0x00, 0x40, 0x7b, 0x3b, 0x7b, 0x7f, 0x55, 0x79, 0x7c, 0x48, 0x58}, // fajar shuru
  {0x20, 0xff, 0xa0, 0xb1, 0xb9, 0xae, 0xa4, 0xa2, 0x3f, 0xa0, 0xff, 0xa0, 0xbf, 0xa1, 0xe5, 0xa6, 0xa0, 0x80,
0x20, 0x34, 0x36, 0x3e, 0x21, 0x20, 0x2f, 0x35, 0x27, 0x20, 0x2e, 0x2b, 0x31, 0x3f}, // ct hobe
  {0x00, 0x4c, 0x5e, 0x72, 0x63, 0x7f, 0x50, 0x5c, 0x40, 0x7f, 0x40, 0x4c, 0x48, 0x4c, 0x48, 0x7f, 0x00, 0x00,
0x4c, 0x5c, 0x72, 0x7f, 0x40, 0x58, 0x53, 0x5d, 0x7f, 0x48, 0x4f, 0x40, 0x00, 0x00}, // kal bondho
  {0x00, 0x00, 0x00, 0xe0, 0xd6, 0x1f, 0x39, 0x31, 0x3f, 0x20, 0x2e, 0x28, 0x28, 0x3f, 0x20, 0x33, 0x3e, 0x20,
0x2c, 0x3a, 0x3f, 0x20, 0x3f, 0x20, 0x3c, 0x24, 0x28, 0x2f, 0x00, 0x00, 0x00, 0x00} // thank you
};
const int numMessages = sizeof(banglaMessagesBitmaps) / sizeof(banglaMessagesBitmaps[0]);

// ── Convert Gregorian → Bangla ──
BanglaDate gregorianToBangla(DateTime g) {
  /*BanglaDate b;
  int day   = g.day();
  int mon   = g.month();
  int yr    = g.year() - 594;
  if (mon > 4 || (mon == 4 && day >= 14)) yr++;
  int m = (mon + 8) % 12;
  if (mon == 4 && day < 14) m = 11;
  int d = day - 13;
  if (d < 1) d += 30;
  if (mon == 4 && day >= 14) { m = 0; d = day - 13; }
  b.day = d; b.month = m; b.year = yr;
  return b;
 */
 BanglaDate banglaDate;

  // Simple approximation (adjust as needed)
  int day = g.day();
  int month = g.month();
  int year = g.year() - 594;

  // Adjust year if after April (Bangla new year is in April)
```

```
  if (month > 4 || (month == 4 && day >= 14)) {
    year++;
  }

  // Check if Gregorian year is leap year (for Falgun correction)
  bool isLeapYear = (g.year() % 4 == 0 && (g.year() % 100 != 0 || g.year() % 400 == 0));

  // Calculate Bangla month (approximate)
  int banglaMonth = (month + 8) % 12;
  if (month == 4 && day < 14) {
    banglaMonth = 11; // Still Chaitro
  }

  // Calculate Bangla day (simple approximation)
  int banglaDay = day - 15;

  // Falgun has 30 days in leap year
  if (banglaMonth == 10 && banglaDay > 29 && !isLeapYear) {
    banglaDay = 1;
    banglaMonth = 11;  // Move to Chaitro
  }

  if (banglaDay < 1) {
    banglaDay += 30;
    // Special case: Falgun in leap year has 30 days
    if (banglaMonth == 10 && isLeapYear) {
      banglaDay += 1;
    }
  }

  // Handle Boishakh starting on 14th April
  if (month == 4 && day >= 14) {
    banglaMonth = 0;
    banglaDay = day - 13;
  }

  banglaDate.day = banglaDay;
  banglaDate.month = banglaMonth;
  banglaDate.year = year;

  return banglaDate;
}

// ── Write a column to chained modules ──
inline void writeColumn(MD_MAX72XX& mx, uint16_t col, uint8_t data) {
  uint8_t dev    = col / 8;
  uint8_t devCol = col % 8;
  mx.setColumn(dev, devCol, data);
}

// ── Display helpers ──
void displayBanglaDigit(MD_Parola &disp, int digit, uint16_t startCol) {
  MD_MAX72XX* mx = disp.getGraphicObject();
  for (uint8_t c = 0; c < 8; c++) writeColumn(*mx, startCol + c, banglaDigits[digit][c]);
}
void displayBanglaMonth(MD_Parola &disp, int month) {
  MD_MAX72XX* mx = disp.getGraphicObject();
  for (uint16_t c = 0; c < DISPLAY_WIDTH; c++) writeColumn(*mx, c, banglaMonthsBitmaps[month][c]);
}
void displayBanglaWeekday(MD_Parola &disp, int wd) {
  MD_MAX72XX* mx = disp.getGraphicObject();
  for (uint16_t c = 0; c < DISPLAY_WIDTH; c++) writeColumn(*mx, c, banglaWeekdaysBitmaps[wd][c]);
}
void displayBanglaMessage(MD_Parola &disp, int msgIdx, int startCol) {
  MD_MAX72XX* mx = disp.getGraphicObject();
  for (uint16_t c = 0; c < DISPLAY_WIDTH; c++) writeColumn(*mx, startCol + c, banglaMessagesBitmaps[msgIdx][c]);
}

// ── URL-decode helper ──
String urlDecode(const String &s) {
  String out; char c;
  for (uint16_t i = 0; i < s.length(); i++) {
    if (s[i] == '+') out += ' ';
    else if (s[i] == '%' && i + 2 < s.length()) {
      String h = s.substring(i+1, i+3);
      c = (char) strtol(h.c_str(), NULL, 16);
      out += c; i += 2;
    } else out += s[i];
  }
  return out;
}

// ── Update time once/sec ──
void updateTimeDisplay(DateTime now) {
```

```
  static unsigned long last = 0;
  if (millis() - last > 1000) {
    last = millis();
    timeDisplay.displayClear();
    int h = now.hour(), m = now.minute();
    displayBanglaDigit(timeDisplay, h/10,  0);
    displayBanglaDigit(timeDisplay, h%10,  8);
    displayBanglaDigit(timeDisplay, m/10, 16);
    displayBanglaDigit(timeDisplay, m%10, 24);
  }
}

// ——— Update date scrolling ———
void updateDateDisplay(BanglaDate bd) {
  static unsigned long last = 0;
  static int pos = 0;
  const unsigned long INT = 150;
  const int GFX = 16 + DISPLAY_WIDTH + 32;
  if (millis() - last > INT) {
    last = millis();
    pos = (pos + 1) % GFX;
    MD_MAX72XX* mx = dateDisplay.getGraphicObject();
    mx->clear();
    int x = -pos;
    displayBanglaDigit(dateDisplay, bd.day / 10, x);
    displayBanglaDigit(dateDisplay, bd.day % 10, x + 8);
    x += 16;
    for (uint16_t c = 0; c < DISPLAY_WIDTH; c++)
      writeColumn(*mx, x + c, banglaMonthsBitmaps[bd.month][c]);
    x += DISPLAY_WIDTH;
    displayBanglaDigit(dateDisplay, (bd.year/1000)%10, x);
    displayBanglaDigit(dateDisplay, (bd.year/100)%10,  x+8);
    displayBanglaDigit(dateDisplay, (bd.year/10)%10,   x+16);
    displayBanglaDigit(dateDisplay, bd.year%10,        x+24);
  }
}

void setup() {
  Serial.begin(115200);
  Wire.begin(21,22);
  if (!rtc.begin()) { Serial.println("RTC missing!"); while (1); }

  // Wi-Fi
  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
  Serial.print("Wi-Fi...");
  while (WiFi.status() != WL_CONNECTED) { delay(300); Serial.print('.'); }
  Serial.println("\nIP: " + WiFi.localIP().toString());
  server.begin();

  // NTP → RTC
  timeClient.begin();
  timeClient.update();
  rtc.adjust(DateTime(timeClient.getEpochTime()));
  lastNtpSync = millis();

  // Initialize matrices
  timeDisplay.begin();  timeDisplay.setIntensity(5);  timeDisplay.displayClear();
  dateDisplay.begin();  dateDisplay.setIntensity(5);  dateDisplay.displayClear();
  msgDisplay.begin();   msgDisplay.setIntensity(5);   msgDisplay.displayClear();
}

void loop() {
  // —— HTTP handler ——
  WiFiClient client = server.available();
  if (client) {
    String req = client.readStringUntil('\r'); client.flush();

    if (req.startsWith("GET /setIndex?idx=")) {
      int p = req.indexOf("idx=")+4, q = req.indexOf(' ',p);
      selectedMessage = req.substring(p,q).toInt();
      useCustomMessage = false;
      msgScrollPos = 0;
      msgDisplay.getGraphicObject()->clear();
    }
    else if (req.startsWith("GET /setSpeed?speed=")) {
      int p = req.indexOf("speed=")+6, q = req.indexOf(' ',p);
      long v = req.substring(p,q).toInt();
      msgScrollInterval = constrain(v, 10, 200);
      if (useCustomMessage) {
        msgDisplay.displayClear();
        msgDisplay.displayText(
          customMessage.c_str(),
          PA_LEFT,
          msgScrollInterval,
```

```
        msgScrollInterval,
        PA_SCROLL_LEFT,
        PA_SCROLL_LEFT
      );
    }
  }
  else if (req.startsWith("GET /setMessage?msg=")) {
    // ——— Parse incoming message as before ———
    int p = req.indexOf("msg=") + 4;
    int q = req.indexOf(' ', p);
    String payload = urlDecode(req.substring(p, q));
    useCustomMessage = true;              // we're now in "custom message" mode
    msgDisplay.displayClear();

    // ——— Build a list of indices ———
    std::vector<int> indices;
    if (payload.indexOf(',') >= 0) {
      // comma-separated: split into two (or more) parts
      int start = 0;
      while (true) {
        int comma = payload.indexOf(',', start);
        String part = (comma >= 0)
                      ? payload.substring(start, comma)
                      : payload.substring(start);
        indices.push_back(part.toInt());
        if (comma < 0) break;
        start = comma + 1;
      }
    } else {
      // no comma: treat the entire payload as a single number
      indices.push_back(payload.toInt());
    }

    // ——— Display each indexed message in turn ———
    for (int idx : indices) {
      if (idx < 0 || idx >= numMessages) continue;     // bounds check!
      msgDisplay.displayClear();
      displayBanglaMessage(msgDisplay, idx, 0);        // draw at column 0
      msgDisplay.displayAnimate();                      // push it out
      delay(2000);                                      // hold for 2 seconds
    }
  }


  client.print(F(
    "HTTP/1.1 200 OK\r\n"
    "Content-Type: text/html; charset=utf-8\r\n\r\n"
  ));
  client.print(WebPage);
  client.stop();
}

// —— Periodic NTP resync ——
if (millis() - lastNtpSync > NTP_RESYNC_INTERVAL) {
  if (timeClient.update()) {
    rtc.adjust(DateTime(timeClient.getEpochTime()));
    Serial.println("RTC re-synced");
  }
  lastNtpSync = millis();
}

// —— Fetch time & date ——
DateTime now = timeClient.update()
                 ? DateTime(timeClient.getEpochTime())
                 : rtc.now();
BanglaDate bd = gregorianToBangla(now);

// —— Update displays ——
updateTimeDisplay(now);
updateDateDisplay(bd);

// —— Message display ——
if (useCustomMessage) {
  if (!msgDisplay.displayAnimate())
    msgDisplay.displayReset();
} else {
  const int GFX = DISPLAY_WIDTH * 2;
  if (millis() - msgLastScroll > msgScrollInterval) {
    msgLastScroll = millis();
    msgScrollPos  = (msgScrollPos + 1) % GFX;
    MD_MAX72XX* mx = msgDisplay.getGraphicObject();
    mx->clear();
    int x = DISPLAY_WIDTH - msgScrollPos;
    displayBanglaMessage(msgDisplay, selectedMessage, x);
```

```
    }
  }

  // Animate Parola
  timeDisplay.displayAnimate();
  dateDisplay.displayAnimate();
  msgDisplay.displayAnimate();
}
```